

Solutions to Problem Set 4

1. Here we use $f_{,1}$ and $f_{,2}$ for the partial derivatives of f with respect to the first and second arguments. We have

$$f_{,1}(t, x) = -\sqrt{x}e^{-t\sqrt{x}}\ln(1+x^2),$$

$$f_{,2}(t, x) = e^{-t\sqrt{x}}\left(\frac{2x}{1+x^2} - \frac{t\ln(1+x^2)}{2\sqrt{x}}\right).$$

By the chain rule, we have

$$\frac{d}{dt}f(t, x(t)) = f_{,1}(t, x(t)) + f_{,2}(t, x(t))x'(t).$$

Substituting in the given values, we find that the value $\frac{d}{dt}f(t, x(t))$ when $t = 3$ is given by

$$-e^{-6}\left[\frac{\ln(17)}{2} + \frac{16}{17}\right].$$

2. By the chain rule, we have

$$\frac{dz}{dt} = 3x^2\frac{dx}{dt} + 2y\frac{dy}{dt}.$$

Applying the chain rule and the product rule, we find that

$$\frac{d^2z}{dt^2} = 3x^2\frac{d^2x}{dt^2} + 6x\left(\frac{dx}{dt}\right)^2 + 2y\frac{d^2y}{dt^2} + 2\left(\frac{dy}{dt}\right)^2.$$

Substituting in the given values, we find that when $t = 3$ the value of $\frac{d^2}{dz^2}$ is given by 15.

3. (a) $\int_0^1 \int_{x^2}^x f(x, y) dy dx$
(b) $\int_1^{e^4} \int_{\frac{1}{2}\ln y}^2 f(x, y) dx dy$
(c) $\int_0^4 \int_0^{\frac{1}{2}y} f(x, y) dx dy$
(d) $\int_0^{\ln 3} \int_{e^y}^3 f(x, y) dx dy$

4. (a)

$$\begin{aligned}\int_0^1 \int_0^{x^3} 4x^2 y \, dy \, dx &= \int_0^1 2x^2 y^2 \Big|_0^{x^3} \, dx \\ &= \int_0^1 2x^2 \cdot x^6 \, dx = \int_0^1 2x^8 \, dx = \frac{2}{9}.\end{aligned}$$

(b) We interchange the order of integration.

$$\begin{aligned}\int_0^1 \int_x^1 3x \sqrt{x^2 + y^2} \, dy \, dx &= \int_0^1 \int_0^y 3x \sqrt{x^2 + y^2} \, dx \, dy \\ &= \int_0^1 (x^2 + y^2)^{\frac{3}{2}} \Big|_0^y \, dy = \int_0^1 [(2y^2)^{\frac{3}{2}} - (y^2)^{\frac{3}{2}}] \, dy \\ &= \int_0^1 [(2)^{\frac{3}{2}} - 1] y^3 \, dy = \frac{1}{4} [(2)^{\frac{3}{2}} - 1].\end{aligned}$$

5.

$$\begin{aligned}\int_R \int xy \, dA &= \int_0^1 \int_y^{2-y} xy \, dx \, dy = \int_0^1 \frac{1}{2} x^2 y \Big|_y^{2-y} \, dy \\ &= \frac{1}{2} \int_0^1 [(2-y)^2 y - y^3] \, dy = \frac{1}{2} [(4-4y+y^2)y - y^3] \, dy \\ &= \frac{1}{2} \int_0^1 [4y - 4y^2] \, dy = \frac{1}{2} (2y^2 - \frac{4}{3} y^3) \Big|_0^1 = \frac{1}{3}.\end{aligned}$$

6. Changing to polar coordinates, we find that

$$\begin{aligned}\int_R \int (x^2 + y^2)^{\frac{3}{2}} \, dA &= \int_0^{2\pi} \int_1^2 r^4 \, dr \, d\theta \\ &= \int_0^{2\pi} \frac{1}{5} r^5 \Big|_1^2 \, d\theta = \frac{2\pi}{5} (2^5 - 1) = \frac{62\pi}{5}.\end{aligned}$$

7. (a) We need to calculate

$$I = \int_{-\infty}^{\infty} e^{2x} \psi(x) \, dx.$$

We can get the answer quickly using moment generating functions. I will actually calculate the integral first and then give a solution using the moment generating function. Recall that

$$\psi(x) = \frac{1}{\nu} \phi\left(\frac{x - \mu}{\nu}\right),$$

where ϕ is the standard normal density function. Making the change of variable

$$y = \frac{x - \mu}{\nu},$$

we find that

$$I = \int_{-\infty}^{\infty} \exp[2\mu + 2\nu y] \phi(y) dy,$$

so that

$$I = \frac{e^{2\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp[2\nu y - \frac{1}{2}y^2] dy.$$

[Here we have made use of the specific form of ϕ .] Completing the square, we see that

$$2\nu y - \frac{1}{2}y^2 = -\frac{1}{2}(y - 2\nu)^2 + 2\nu^2.$$

We conclude that

$$I = \frac{e^{2\mu+2\nu^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y-2\nu)^2} dy.$$

Making the substitution $z = y - 2\nu$, we see that

$$I = \frac{e^{2\mu+2\nu^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz = e^{2\mu+2\nu^2}.$$

Remark: We can get the value immediately from the moment generating function m for a normal RV with mean μ and variance ν^2 . Recall that

$$m(\tau) = e^{\mu\tau + \frac{1}{2}\nu^2\tau^2}.$$

Observe that

$$I = m(2) = e^{2\mu+2\nu^2}.$$

The option price is given by

$$V_0 = e^{-rT} I = e^{-rT} \exp[2 \ln S_0 + 2rT - \sigma^2 T + 2\sigma^2 T] = e^{rT} e^{\sigma^2 T} (S_0)^2.$$

(b) We see that

$$\Delta_0 = 2e^{rT} e^{\sigma^2 T} S_0.$$